23CSE202 DATABASE MANAGEMENT SYSTEMS

S3 B.Tech CSE
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

2. UI Design Document

Group No : D9

Team Members :

Roll No	Student Name
AM.SC.U4CSE25359	Abel Binu Varghese
AM.SC.U4CSE25360	Alien Lakshmi C
AM.SC.U4CSE25366	Kasinath V
AM.SC.U4CSE25368	Devananda J A

Project Title : Contact Management System

2.1. Functional requirements

The main functionalities required for the UI are:

1. User authentication with a Sign In / Sign Up flow (default admin/admin123 account); new self-registered accounts are created only as Editor or Viewer.
2. Add, view, edit, delete and search contact records (name, phone, email, address, birthday, company, category), with duplicate phone/email detection.
3. Categorize contacts (Personal, Business, Family, Academic), mark/filter favourites, jump to a letter via an A-Z rail, and sort by Name A-Z/Z-A, Recently added, or Company.
4. Role-based access control for Administrator, Editor and Viewer roles that shows or hides add/edit/delete and admin actions accordingly.
5. Admin-only Manage Users panel to add new users with a chosen role, and an Activity Log panel showing logins, logouts and contact/user changes with timestamps.
6. Confirmation popup before any delete (contact or user) to prevent accidental data loss.
7. Dashboard summary cards showing total contacts, number of categories, companies and favourites.

2.2. Non functional requirements

The main qualitative features required are:

1. Persistence without a server: all accounts, contacts and activity logs are stored in the browser's LocalStorage and survive a page refresh.
2. Usability: a clean, distraction-free record-drawer themed interface with clear inline validation messages for invalid phone numbers or email addresses.
3. Performance: near-instant response for search, filter, sort and CRUD operations since all data is held and processed client-side.
4. Security: passwords are hashed (non-cryptographic, classroom-appropriate) before storage, and UI actions are restricted according to the logged-in user's role.
5. Maintainability: the UI logic, styling and markup are separated into distinct app.js, styles.css and index.html files for clarity.
6. Visual consistency: a moody near-black theme with off-white cards, a serif heading font, a monospace label font, and muted steel-blue/slate accents throughout.
7. Responsiveness: a collapsible sidebar and adaptive grid so the layout works across desktop and mobile browsers.

2.3. List of AI tools used for building UI (if any)

1. Claude (Anthropic) - used to generate and refine the HTML/CSS/JS for the UI.

2.4. Prompt (if any)

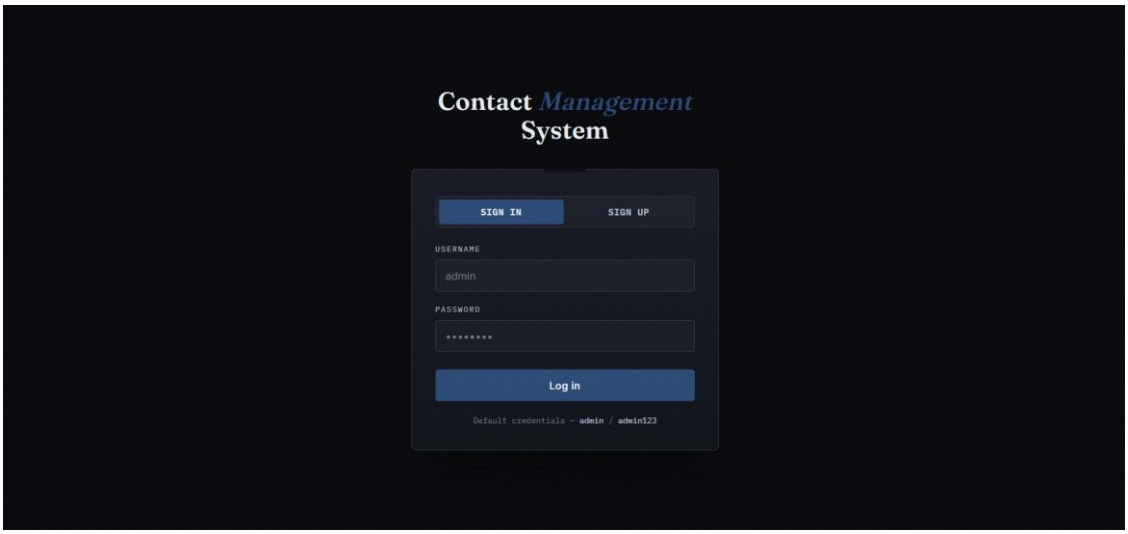
Sl. No	Name of the tool	Prompts
1	Claude (Anthropic)	Build a dark, "record drawer"-themed contact management website using plain HTML, CSS, and JavaScript only, with all data (accounts, contacts, activity log) saved in the browser's local storage so nothing needs a real server. Start users on a login screen with Sign In/Sign Up tabs (default admin/admin123, new signups become Editor or Viewer only), then show a sidebar with categories, an A-Z letter filter, and admin-only "Manage users" and "Activity log" buttons, plus a top bar with search, sorting, a favorites toggle, and an "Add contact" button. Display contacts as cards in a grid (name, company, phone, email, category tag) that open into a detail view, with edit/add/delete forms validating a 10-digit phone and proper email, and a confirmation popup before any delete. Support three roles - Administrator, Editor, Viewer - that show or hide edit/delete/manage-user abilities accordingly. Use a moody near-black background, off-white cards, a serif heading font, a monospace label font, and muted steel-blue/slate accent colors throughout.

2.5. List of files

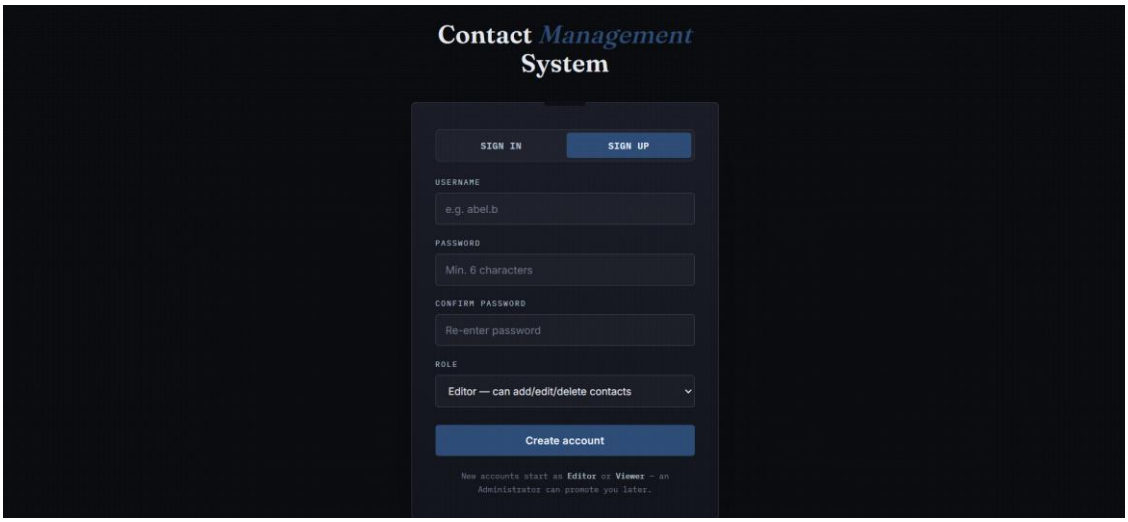
HTML / CSS / JS files created for the UI:

Sl. No	File name
1	index.html
2	styles.css
3	app.js

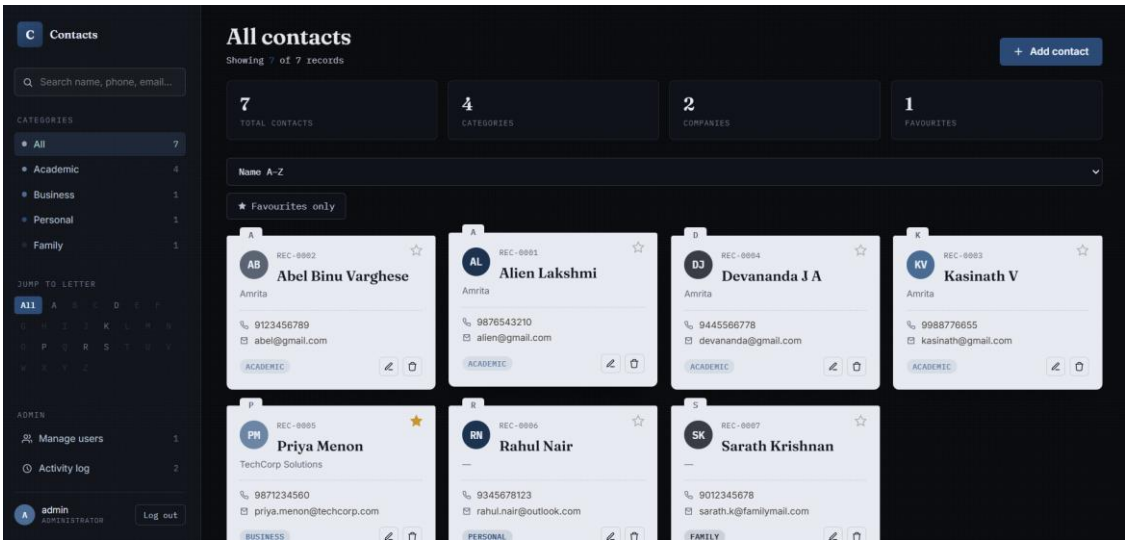
2.6. UI Screenshot



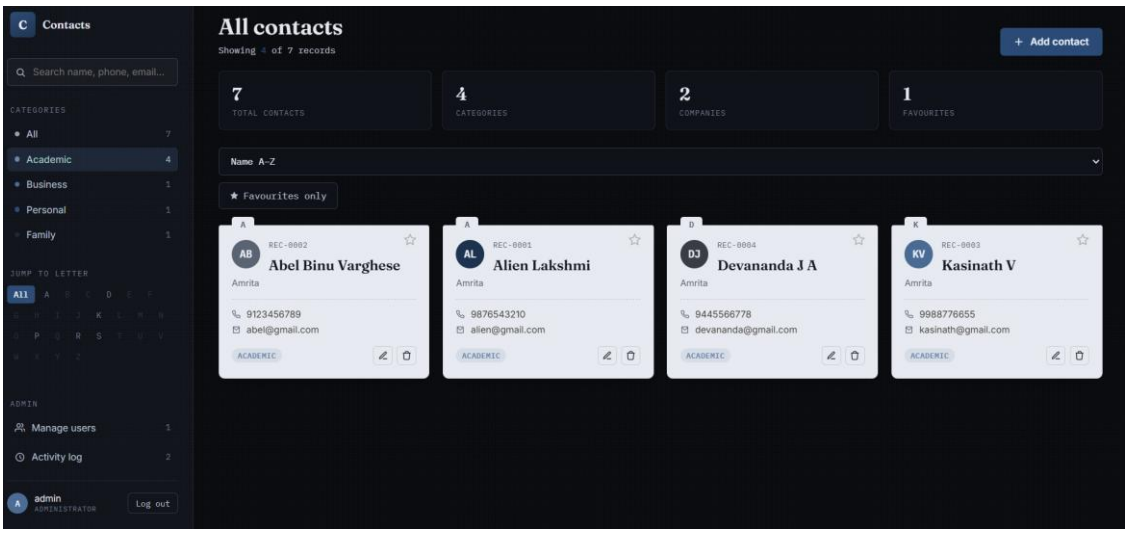
Sign in screen



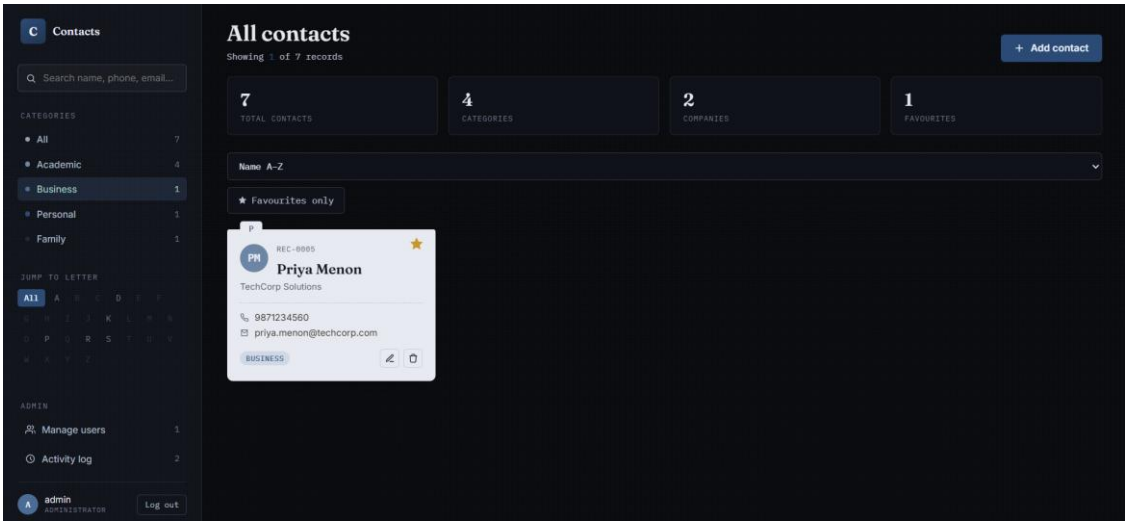
Sign up screen



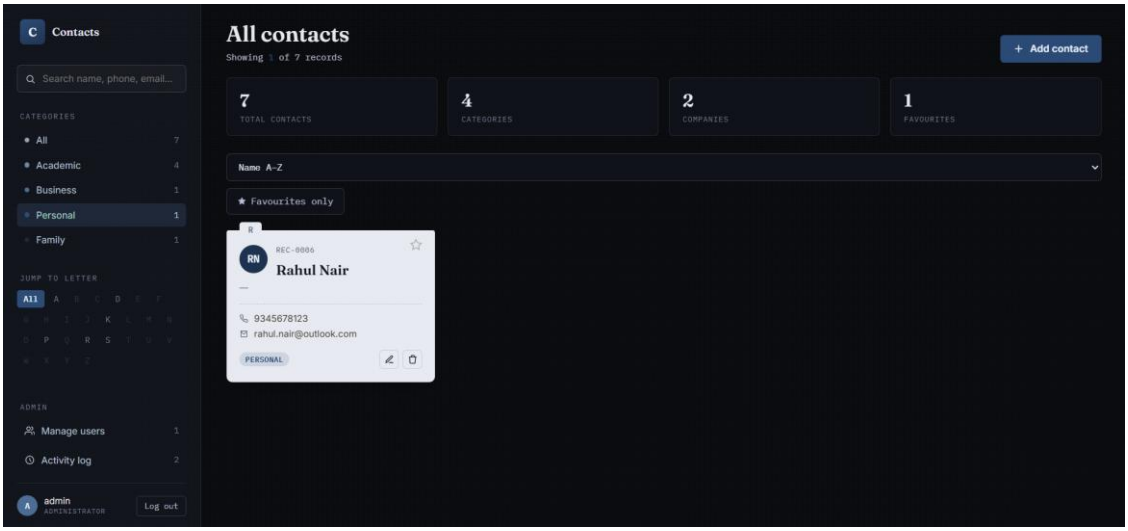
All contacts - dashboard and card grid



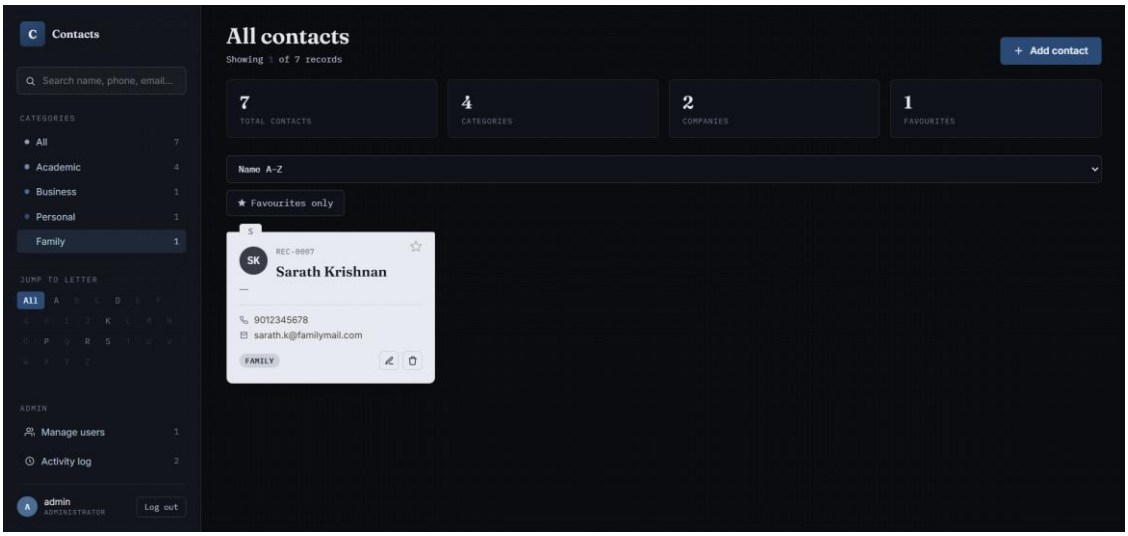
Contacts filtered by Academic category



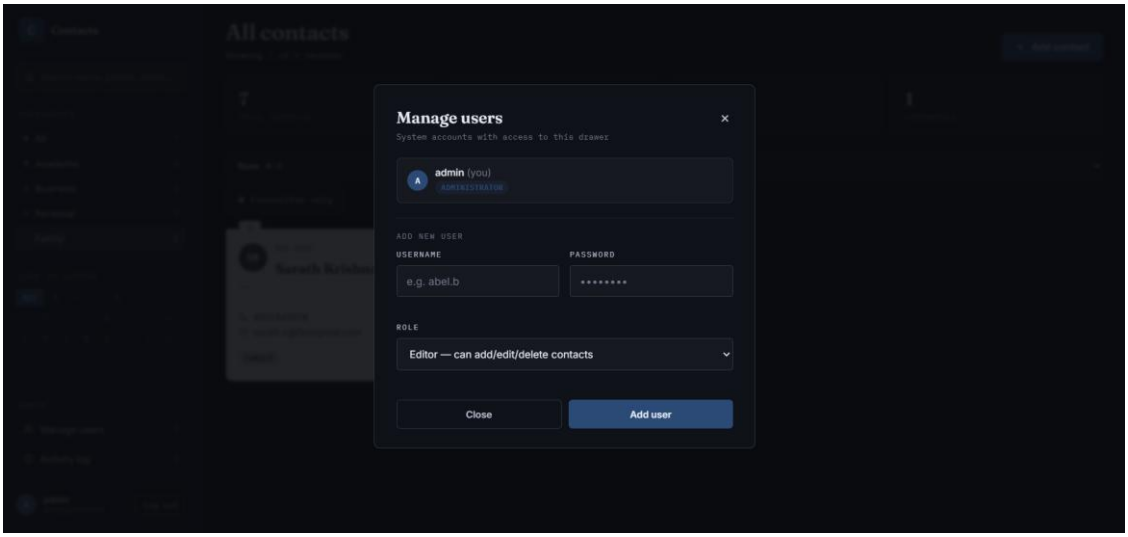
Contacts filtered by Business category



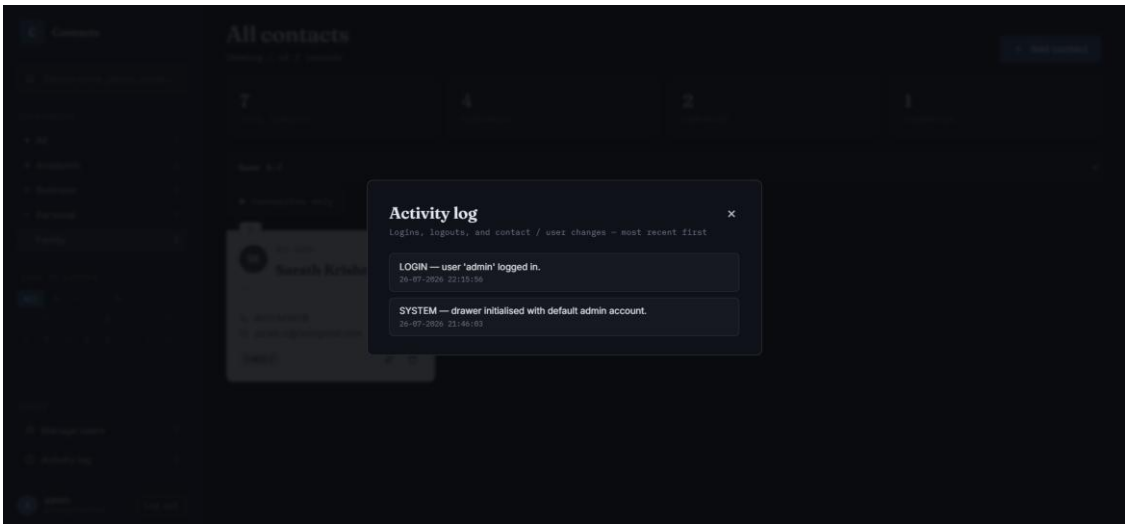
Contacts filtered by Personal category



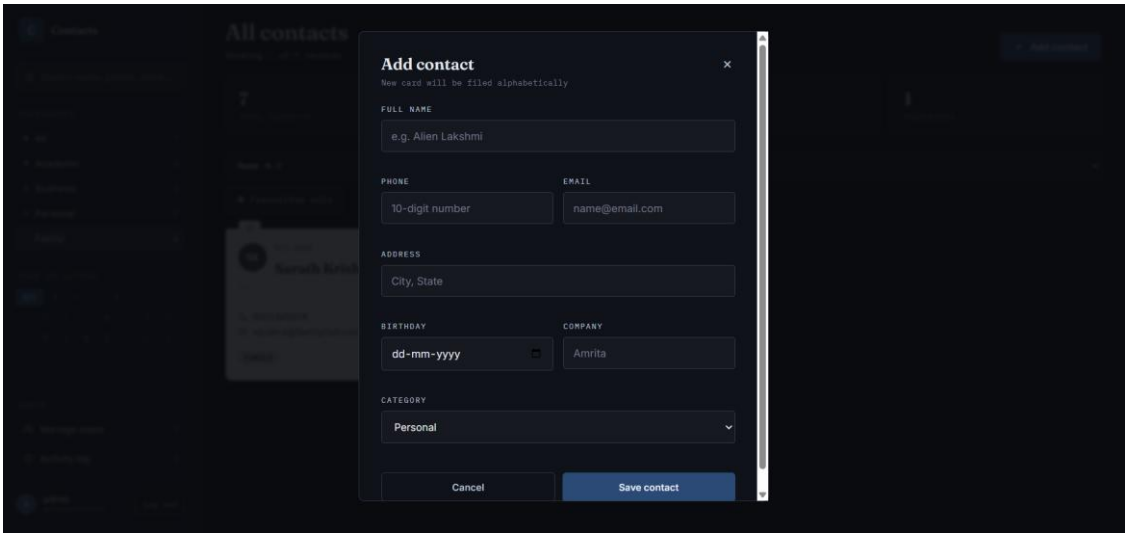
Contacts filtered by Family category



Manage users modal (Admin)



Activity log modal (Admin)



Add contact form

2.7. GitHub Link

https://alien2088.github.io/23CSE202_ContactManagementSystem_D9/